

FLOATING POINT REPRESENTATION

PRACTICE SHEET • EASY / SET 2

ResourceOS

1. In IEEE 754 floating-point format, what does a sign bit value of 1 indicate?

- (A) Negative number
- (B) Positive number
- (C) Subnormal number
- (D) Division by zero error

2. How many bits are dedicated to the fraction (mantissa) in IEEE 754 double-precision format?

- (A) 52 bits
- (B) 23 bits
- (C) 11 bits
- (D) 64 bits

3. How many bits are allocated for the exponent field in IEEE 754 double-precision format?

- (A) 11 bits
- (B) 8 bits
- (C) 15 bits
- (D) 32 bits

4. What is the binary value of decimal fraction 0.125?

- (A) 0.001_2
- (B) 0.01_2
- (C) 0.101_2
- (D) 0.011_2

5. If the actual exponent of a floating-point number is 0, what is the stored exponent field value in IEEE 754 single precision?

- (A) 127
- (B) 0
- (C) 128
- (D) 255

6. Express decimal 2.0 in normalized binary scientific notation ($1.M \times 2^E$).

- (A) $1.0_2 \times 2^1$
- (B) $0.1_2 \times 2^2$
- (C) $10.0_2 \times 2^0$
- (D) $1.0_2 \times 2^2$

7. What is the formula to convert an unbiased exponent E into a stored exponent e in single precision?

- (A) $e = E + 127$
- (B) $e = E - 127$
- (C) $e = E + 1023$
- (D) $e = 127 - E$

8. What is the implicit normalized bit assumed to exist to the left of the binary point in IEEE 754 standard?

- (A) 1
- (B) 0
- (C) Sign bit value
- (D) Parity bit

9. Which floating-point field is placed at the most significant bit (MSB) position in standard formats?

- (A) Sign bit
- (B) Exponent
- (C) Mantissa
- (D) Guard bit

10. Convert decimal fractional number 0.375 to binary representation.

- (A) 0.011_2 (B) 0.101_2
(C) 0.110_2 (D) 0.0011_2

11. What actual exponent value corresponds to a stored 8-bit exponent of 129 in IEEE 754 single precision?

- (A) +2 (B) +1
(C) -2 (D) +129

12. What does an IEEE 754 float with exponent = 0 and mantissa = 0 represent?

- (A) Zero (0.0) (B) Infinity
(C) NaN (D) Smallest normalized number

13. Express the decimal number -4.5 in binary scientific notation.

- (A) $-1.001_2 \times 2^2$ (B) $-10.01_2 \times 2^1$
(C) $-0.1001_2 \times 2^3$ (D) $-1.01_2 \times 2^2$

14. What sign bit is used to store decimal 0.0 in floating-point?

- (A) Either 0 (+0) or 1 (-0) (B) Always 0
(C) Always 1 (D) Undefined

15. In IEEE 754 single precision, if the exponent field is 11111111_2 and mantissa is 0 with sign bit 1, it represents:

- (A) -Infinity ($-\infty$) (B) +Infinity ($+\infty$)
(C) Quiet NaN (D) Overflow limit

16. Which range of stored exponents (in decimal) is reserved for normalized single-precision numbers?

- (A) 1 to 254 (B) 0 to 255
(C) -126 to +127 (D) 1 to 127

17. What is the mantissa bit pattern for decimal 1.0 in single precision?

- (A) All 0s (23 zeros) (B) All 1s (23 ones)
(C) 1000... (1 followed by zeros) (D) 0100...

18. Convert binary fraction 0.101_2 to its decimal equivalent.

- (A) 0.625 (B) 0.5
(C) 0.75 (D) 0.375

19. What is the main advantage of normalizing floating-point numbers?

- (A) It ensures a unique representation for each non-zero number
(B) It doubles the stored precision bits
(C) It prevents overflow completely
(D) It removes the sign bit requirement

20. What is the actual unbiased exponent for decimal 0.25 when normalized?

- (A) -2
(B) -1
(C) 0
(D) +2

21. What is the stored 8-bit exponent field for decimal 0.25 in IEEE 754 single precision?

- (A) 125 (01111101)
(B) 127 (01111111)
(C) 129 (10000001)
(D) 123 (01111011)

22. In IEEE 754 single precision, what does the exponent value 00000000_2 combined with a non-zero mantissa represent?

- (A) Subnormal (Denormalized) number
(B) Overflow error
(C) Normal number
(D) Signalling NaN

23. Convert decimal fraction 0.0625 into powers of 2.

- (A) 2^{-4}
(B) 2^{-3}
(C) 2^{-5}
(D) 2^{-2}

24. Which bit sequence represents decimal fraction 0.0625 in binary notation?

- (A) 0.0001_2
(B) 0.0010_2
(C) 0.0100_2
(D) 0.00001_2

25. In single precision floating-point, how many decimal digits of approximate precision are provided by 23 mantissa bits?

- (A) 6 to 7 digits
(B) 15 to 17 digits
(C) 3 to 4 digits
(D) 23 digits

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)